. Worksheet No. 1

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EXPLORING QUADRATIC FUNCTIONS OF THE FORM $y = ax^2 + bx + c$

Group:.....

Read carefully and fill in the blanks:

Task 1: Visualise the curve's shape

Step 1: Scan QR Code below to access GeoGebra Classroom.

Step 2: On the left panel, click the "Algebra" tab (the calculator icon).

Step 3: Click the input bar and type: $y = ax^2 + bx + c$ then press "Enter".

Step 4: Move the sliders for b and c to any value.

Step 5: Now, focus on slider a. Slide it and observe the change:

- When $a > 0$, the graph is $\cup$-shaped.
- When $a < 0$, the graph is $\cap$-shaped.

Task 2: Consider the graph $y = 2x^2 + 4x - 6$, follow the following steps to find the essential information for sketching the graph:

1. Shape: $a = 2 > 0$, so the graph is $\cup$-shaped.
2. y-intercept: Set $x = 0$ then $y = -6$. Coordinate: (0,-6).
3. x-intercepts: Set $y = 0$ then we obtain $2x^2 + 4x - 6 = 0$.

- Factorize: $2(x-1)(x+3) = 0$.
- $x = 1$ or $x = -3$. Coordinates: (1, 0) and (-3, 0).

4. Turning Point:

- Find axis of symmetry: $x = -\frac{b}{2a} = -\frac{4}{2.2} = -1$.
- Substitute $x = -1$ to find $y = 2.(-1)^2 + 4.(-1) - 6 = -8$.
- Turning point coordinate: $(-1, -8)$.

1. Correct your answer based on the AI's feedbacks.

Task 3: Find the turning point by completing the square

You can find the coordinates of the turning point of a parabola algebraically by changing the equation from $y = ax^2 + bx + c$ *into the form* $y = a(x+p)^2 + q$. *In this form, the turning point is* $(-p, q)$.

For example, let us consider the equation: $y = 2x^2 + 8x - 8$.

Method 1: By formula

Step 1: Identify the coefficients from the standard form $y = ax^2 + bx + c$:

$a = 2$, $b = 8$, $c = -8$.

Step 2: Find the x-coordinate using the formula $x = -\frac{b}{2a}$: $x = -\frac{8}{2.2} = -2$.

Step 3: Find the y-coordinate by substituting the value of x into the original equation: $y = 2.(-2)^2 + 8.(-2) - 8 = -16$.

Step 4: Based on calculation, the turning point is: (-2,-16).

Method 2: By completing the square

Consider the equation $y = 2x^2 + 8x - 8$

This can be rewritten as $y = 2(x+2)^2 - 16$ by completing the square.

Squaring any value results in an answer that is either positive or 0. This means that for any value of x, the smallest value of $(x+2)^2$ is 0.

This means that the minimum value of $y = 2(x+2)^2 - 16$ is -16 and that this occurs when $x = -2$.

The turning point of the graph $y = 2(x+2)^2 - 16$ has the coordinates: $(-2, -16)$.